Tekendra Bhatt, Ph.D.

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Education

- **Ph.D. in Mathematics (Probability & Stochastic Processes)** 2026
  University of North Texas, Denton, TX
  Dissertation: Estimation and Statistical Inference for Hawkes Processes
  Advisor: Dr. Kai-Sheng Song.
- **M.A. in Economics** 2012
  Kumaun University, Nainital, India.
- **B.Ed. in Mathematics** 2006
  Tribhuvan University, Kathmandu, Nepal.
- **M.A. in Mathematics** 2004
  Tribhuvan University, Kathmandu, Nepal.
- **B.A. in Mathematics & Economics** 2002
  Tribhuvan University, Kathmandu, Nepal.

Academic Appointments and Teaching Experience

Assistant Professor of Mathematics August 2026 - Present
Augusta University, Augusta, GA

- Teach College Algebra and Precalculus, with responsibility for instruction, assessment, grading, and student support.
- Teach College Algebra and Precalculus, with responsibility for instruction, assessment, grading, and student support.
- Scheduled to teach Elementary Statistics (MATH 1401) and Precalculus (MATH 1113) in Spring 2027.
- Incorporate comprehensive group projects as course requirements in both courses to deepen understanding and support collaborative learning.
- Offer individual meetings and maintain an open-door approach to encourage students, address learning difficulties, and support academic progress.
- Use D2L Brightspace and Pearson MyLab Math to organize course materials, assignments, and feedback.

Teaching Fellow / Teaching Assistant 2018 - 2026
University of North Texas, Denton, TX

- **Instructor of Record:** Taught undergraduate Precalculus (in person and online) and Business Calculus; responsible for lectures, assessments, grading, and student support.
- **Teaching Assistant:** Supported large-enrollment Calculus I and II courses by leading small recitation sections (under 40 students) aligned with common lectures, assessments, and course objectives.

- **Student Support:** Provided structured office hours, early outreach to at-risk students, and individualized feedback.
- **Mathematics Assistance Center:** Provided one-on-one instructional support to STEM students in the Mathematics Learning Lab (MathLab), covering courses from Precalculus to Differential Equations.
- **Teaching Recognition:** Recipient of the *Outstanding Teaching Fellow Award* (2023).

Assistant Professor of Mathematics 2015 - 2018
Far Western University, Nepal

- Taught undergraduate and graduate courses, including Calculus, Differential Equations, Linear Algebra, Real Analysis, Topology, Abstract Algebra, Business Mathematics, and Statistics.
- Guided undergraduate course projects designed to reinforce mathematical concepts and deepen student learning.

Mathematics Faculty 2010 - 2015
Brixton College and **Kanchan Vidya Mandir College**, Nepal

- Taught high-school mathematics (Grades 11 - 12)
- Taught Business Mathematics and Business Statistics courses for BBA and BBS programs.
- Instructed large-enrollment courses with approximately 40 - 100 students, emphasizing structured pacing and scalable assessment.

Mathematics Faculty 2005 - 2010
Bal Jagritee College, Nepal

- Taught Calculus, Linear Algebra, and Real Analysis for B.Ed. programs.

Teaching Interests

Core Undergraduate Mathematics:
Calculus I - III, Linear Algebra, Differential Equations, Discrete Mathematics, Business Calculus, Introduction to Proofs.

Probability, Statistics, and Data Science:
Probability, Introductory Statistics, Mathematical Statistics, Statistical Computing with R, Introduction to Data Science, Financial Mathematics.

Upper-Division & Applied Mathematics:
Real Analysis I - II, Abstract Algebra, Complex Analysis, Vector Calculus, Numerical Analysis, Mathematical Modeling.

Additional Teaching Interests:
Course coordination for multi-section courses, project-based learning, undergraduate project mentoring, and interdisciplinary applications.

Research Interests

- **Probability Theory:** Self-exciting point processes, oscillatory Hawkes processes, ergodic theory of stochastic processes, moment and cumulant analysis.

- **Stochastic Processes:** Temporal point processes, limit theorems for dependent data, model stability and inference for complex temporal dependencies.
- **Statistical Inference:** Maximum likelihood estimation for point processes, asymptotic theory, Fisher information analysis, model diagnostics and validation.
- **Computational Statistics:** Efficient simulation algorithms, statistical computing in R, reproducible research workflows, open-source software development.
- **Interdisciplinary Applications:** Temporal modeling in social, environmental, and financial systems.

Papers and Works in Progress

- **Non-monotonic Hawkes Processes: Theory, Inference, and Applications**
  Manuscript in preparation.
- **Analytic Moment Formulas for Hawkes Processes via Differential Operators**
  Manuscript in preparation.
- **Utilizing the Hawkes Process to Apply Orthogonality in Normed Linear Space**
  Manuscript ready for submission.
- **Software Package for Simulation and Inference in Hawkes Processes (R)**
  Simulation, estimation, and diagnostic tools; documentation in progress.

Research Experience

Ph.D. Researcher 2022 - 2026

University of North Texas, Department of Mathematics

- Developed a new class of oscillatory Hawkes processes with sinusoidally modulated, exponentially decaying excitation kernels to model nonmonotonic event dynamics.
- Established theoretical conditions ensuring stability, ergodicity, and identifiability of the proposed models.
- Proved consistency and asymptotic normality of maximum likelihood estimators, with supporting analytic approximations for likelihood derivatives.
- Derived analytic expressions and approximations for the Fisher information matrix, validated through extensive Monte Carlo studies.
- Implemented simulation, estimation, and diagnostic tools in R, enabling reproducible inference and model comparison.
- Applied the models to high-frequency financial event data, demonstrating improved calibration and interpretability compared to classical monotone Hawkes models.
- Developed a unified analytic framework for moment and cumulant calculations applicable to exponential, gamma, and oscillatory Hawkes kernels.

Presentations and Proposed Student Research

When Events Breed Events: A Capstone Project Pitch 2026
Augusta University, presentation to master's students.

- Developed and presented a proposed capstone project on Hawkes process modeling and probabilistic forecasting, with potential applications to neural spike trains and cybersecurity event logs.
- Outlined a semester plan covering simulation, parameter estimation, comparison with a Poisson baseline, forecast evaluation, and a final report and presentation. Student participation is pending.

Professional Service and Leadership

Augusta University 2026 - Present

- **Member, College Algebra and Precalculus Core Committees:** Prepare common examinations, review assessment questions, coordinate course content, and contribute to discussions of assessment policies.
- **Member, CLASS Faculty Liaison Committee:** Collaborate on connecting academic support services with classroom instruction, including early intervention, effective study strategies, and student persistence.
- **Member, Mathematics Contest Committee:** Serve on the committee for the university's annual mathematics contest, with the 52nd contest scheduled for February 2027.

Previous Service

- **Instructional Design (UNT):** Adapted and enhanced department-provided Canvas course shells to support both synchronous and asynchronous delivery; customized weekly modules and assessments to optimize engagement for self-paced and live-remote students.
- **Examination & Assessment Leadership:** Collaborated with the Office of the Controller of Examinations (Far Western University) to prepare and moderate final exam papers and coordinate grading policies across the department.
- **Curriculum Development:** Served on the undergraduate curriculum committee (Far Western University), contributing to the design of the mathematics program and authoring content for the Topology course.
- **University Service:** Served on committees for annual sports and co-curricular activities; hosted student orientation programs and internal workshops.
- **Event Leadership:** Served as Master of Ceremonies for the University Convocation and other major academic events, facilitating large-scale public gatherings.

Awards and Honors

- Outstanding Teaching Fellow Award, Department of Mathematics, UNT 2023
- Nominee, University-wide Outstanding Teaching Fellow Award, UNT 2023
- Academic Excellence Award, Department of Mathematics, UNT 2022

Grants and Scholarships

- Summer Research Assistant Award, College of Science, UNT 2025, 2024
- Texas Public Education Grant-International (TPEG-I), UNT 2024
- Tuition Benefit Program, Graduate School, UNT 2018–2026
- R. Mauldin Scholarship, Department of Mathematics, UNT 2021–2022

Technical & Teaching Tools

- **Programming:** R, LaTeX.
- **Workflows:** Reproducible research with R Markdown.
- **Teaching Technology:** D2L Brightspace, Pearson MyLab Math, Canvas, WebAssign, Respondus, Microsoft Forms, Mathematica, Desmos, GeoGebra.

References

Available upon request.